

Opportunities to Explore

SUGAR Volvo CE
2024/25

PRESENTED BY

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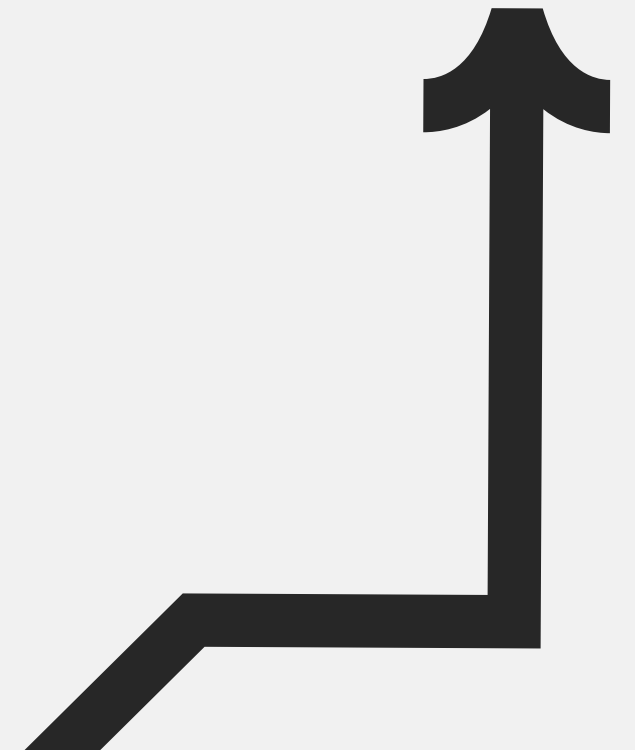
Helena Moyen

Leonardo Valente

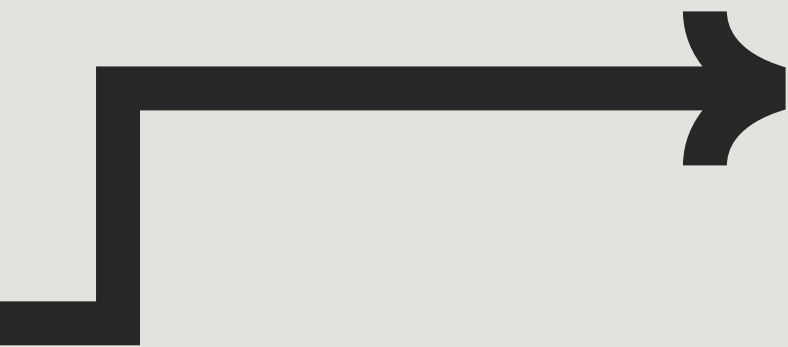
Madhav Sai Aryan

Maja Svensson

Murilo Monteiro



Meeting Agenda



First Opportunity

- **Problem:** Manual Maintenance Analysis - Inefficiency and Limited Integration
- **Solution:** Intelligent Predictive Maintenance - AI-Powered Insights and Integration

Second Opportunity

- **Problem:** Disconnected Workflows - Challenges in Manual Monitoring and Field-Team Integration
- **Solution:** Real-Time Optimization - Intelligent Routing and Operational Insights

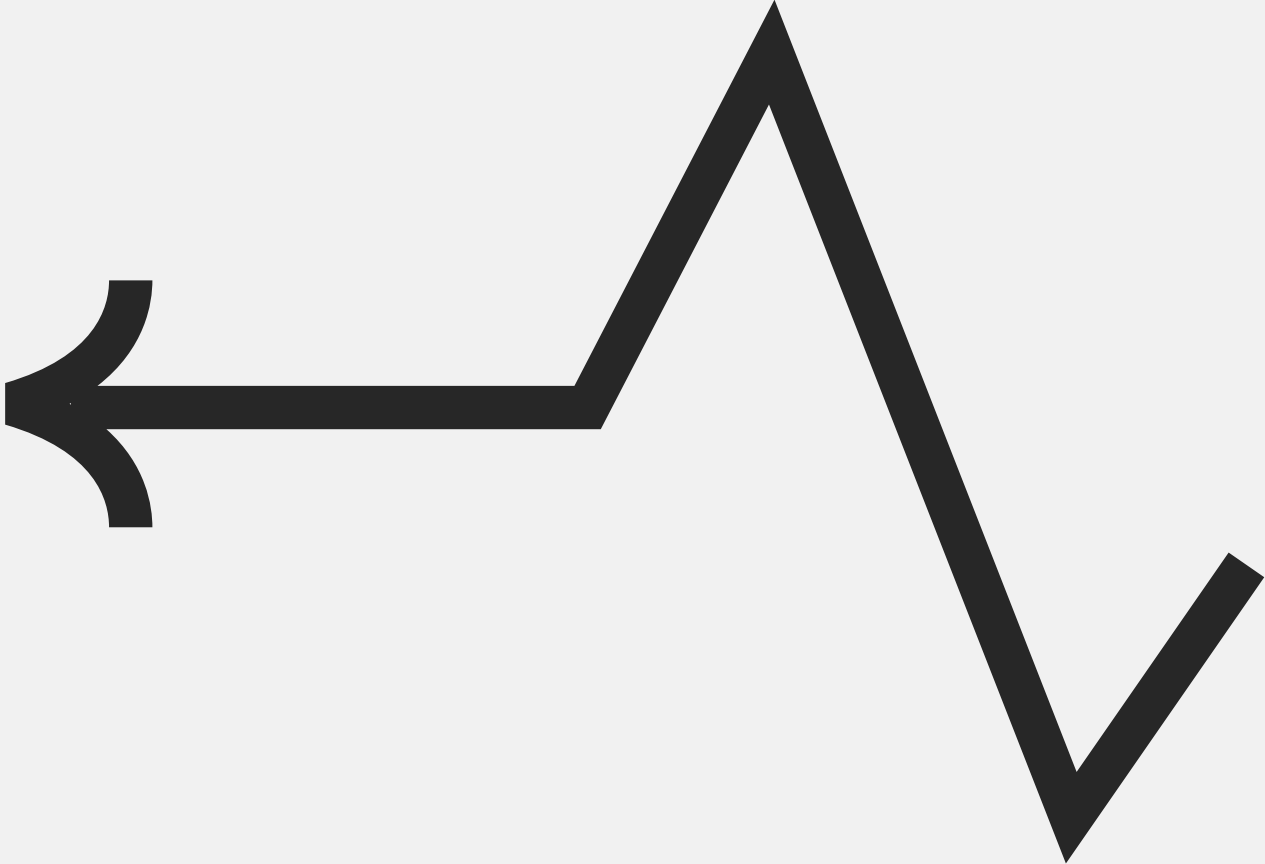
Third Opportunity

- **Problem:** Connectivity Challenges, Data Overload Integration Gaps.
- **Solution:** hybrid Communication Framework-Software defined networking and Edge computing

Feedback (Q&A)

Introduction to the Problem

Manual Maintenance Analysis



About repair prediction:
***"The analysis is not automated;
technicians analyze the data
manually, combining both Oil
Analysis and Care Track data"***

Rodrigo Santos,
Key Account Sales Manager
at Volvo CE



About data interpretation:
***"There is a gap between the
information that Volvo offers
and what the costumer
understands from that"***

José Gabriel Alas Mayer,
Sales Engineer
at Volvo CE



Issues

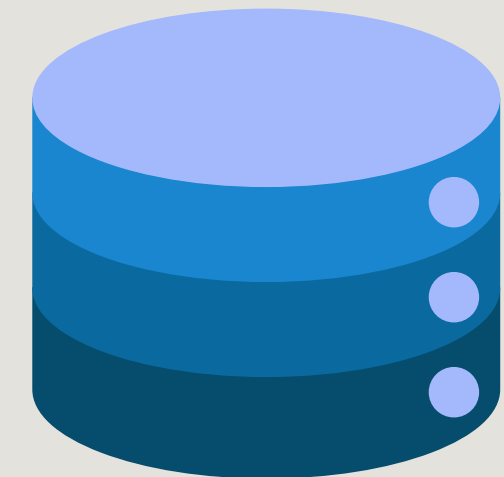
- 01 Customer Interpretation:** Difficulty in handling and analyzing data by themselves



- 02 Lack of Automation:** Analysis is performed manually by technicians using multiple data sources



- 03 Large data volume:** Challenges in managing large data volumes to create reports and perform historical analysis across different contexts



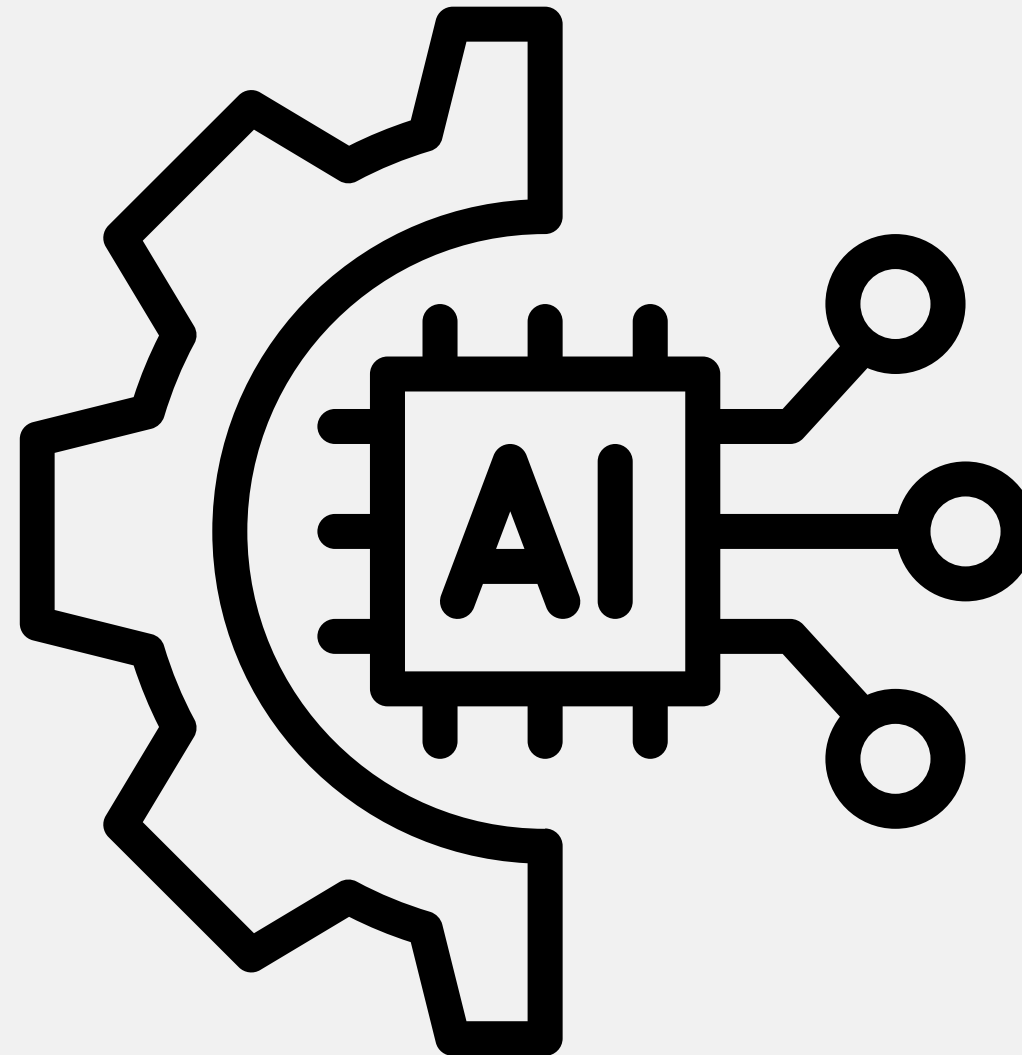
Solution

Integrated database

Combine data from CareTrack, Oil Analysis and critical contextual information (e.g., field of work) .

Complete and automated analysis

Using the integrated and historical data, create analysis and predictions for machines

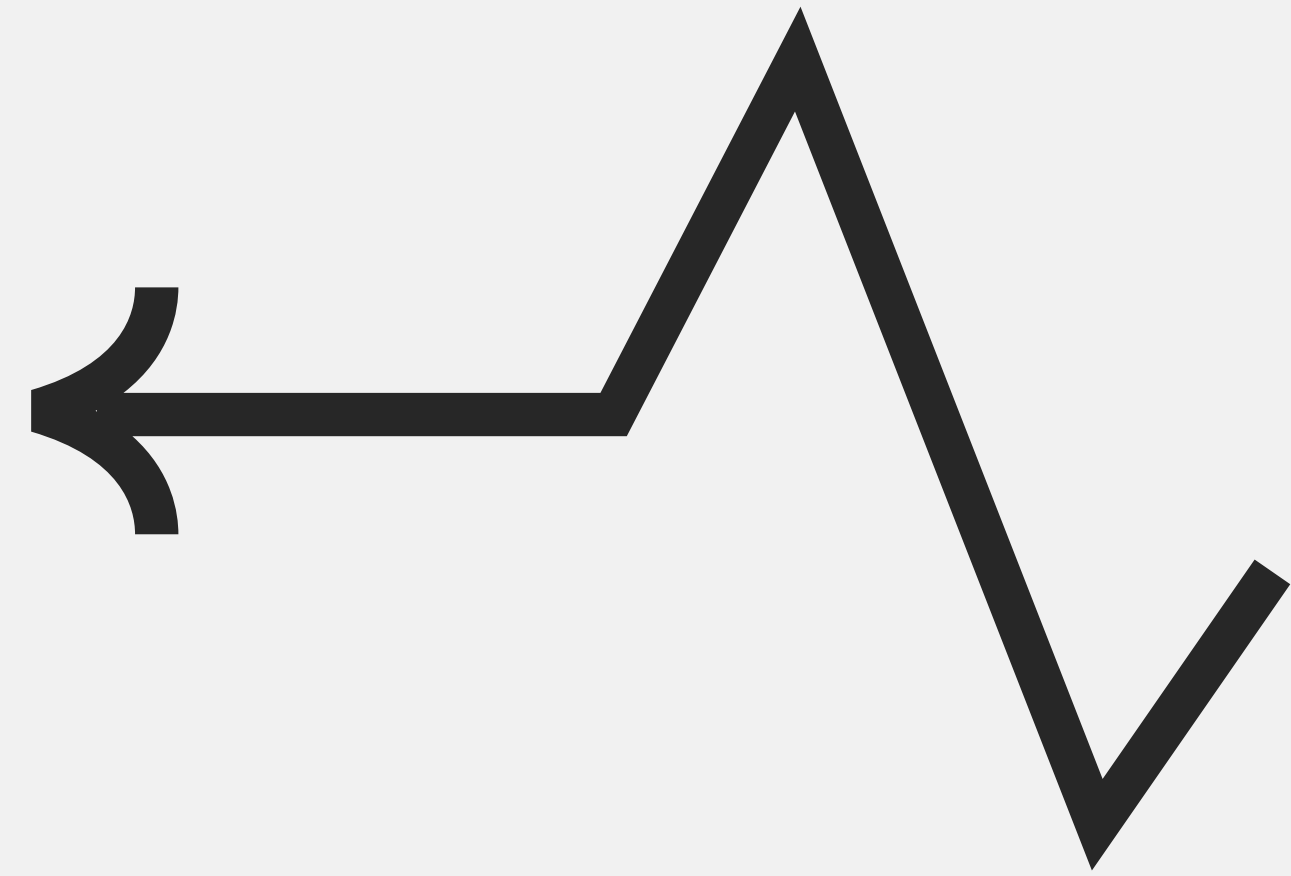


User-Friendly Reports

Automatic creation of easy-to-read, intuitive reports for customers, highlighting relevant data and providing repair/maintenance recommendations

Introduction to the Problem

Manual Construction Site Management



About the Construction Site:

"The construction site is a dynamic entity, where configurations and decisions can constantly change due to both external and internal factors"

Edson Luiz Pechio,
CEO
at Grupo Base



About the management:

"The mine is a long-established business that has been operated in the same way for many years. As a result, there is a tendency to keep decisions in the hands of humans"

Giorgio De Tomi,
Mining Engineering Professor
at USP

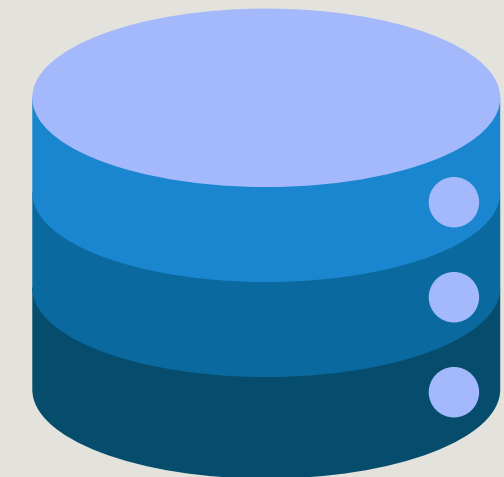


Issues

- 01 Lack of Automation and Intelligence in Operational Monitoring:** Inefficiencies and low velocity of response of changes in operation.



- 02 Fragmented Integration between Field Operators and Central Management:** Decisions are made on site, based on experiences, without the complete analysis of the possibles solutions and data.



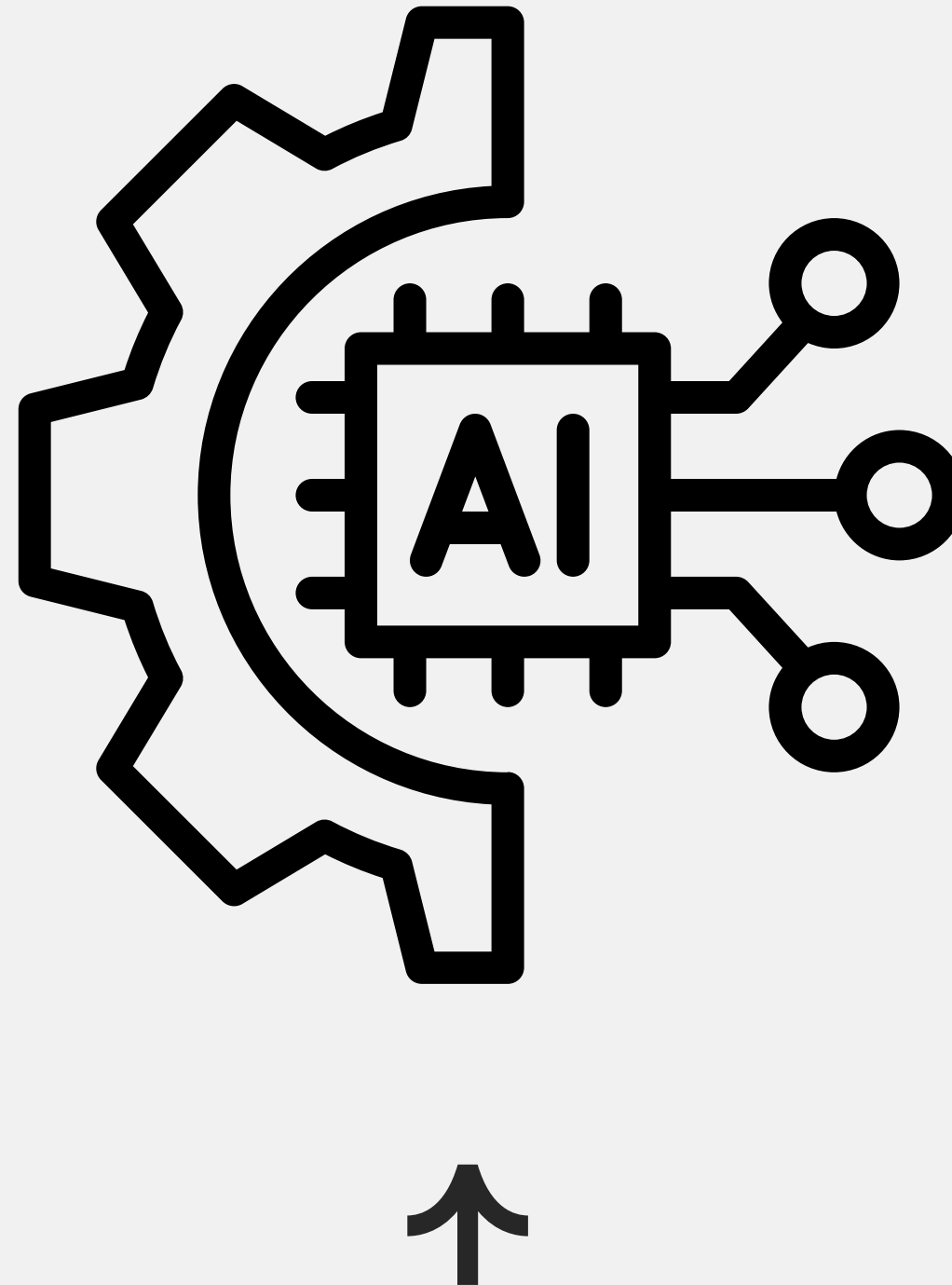
Solution

Integrated database

Combine data from CareTrack and Connected Map

Real-time analysis

Use real-time data for route optimization, anomaly alerts, and the generation of suggestions

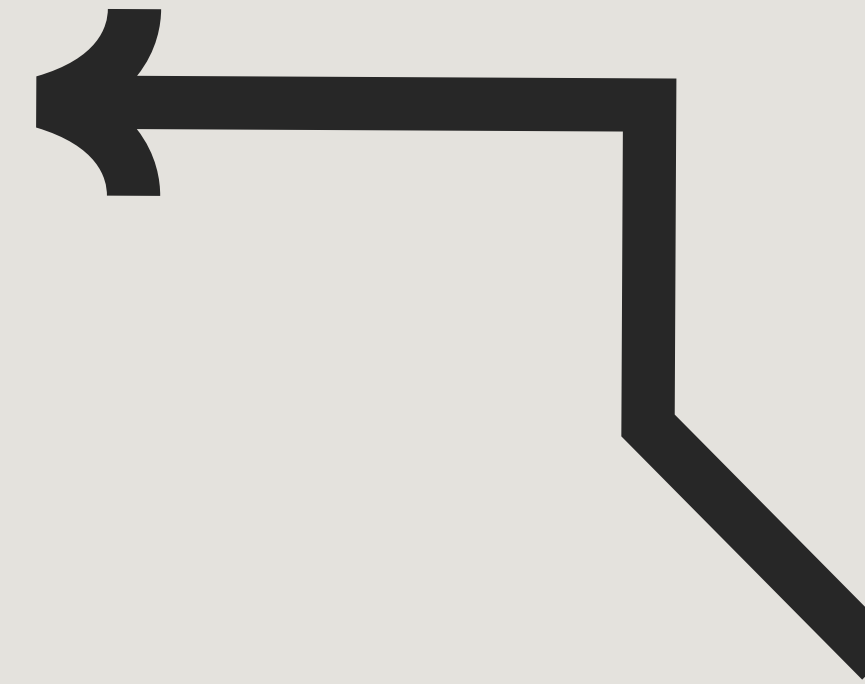


Results

With this solution, the operation would align human decision-making while supporting it with a solid data foundation, reducing downtime and increasing productivity

Challenges expected

- **Data Integration Complexity:** Managing data from multiple sources can be challenging, especially when ensuring compatibility and consistency across systems.
- **Cross-Brand Machine Data Gathering:** Collecting data from machines of different brands may involve dealing with proprietary formats and limited interoperability.
- **Resistance from Small/Medium Customers:** Smaller businesses might hesitate to invest in a comprehensive system due to cost concerns or perceived complexity.



Issues

- 01 Connectivity Challenges:** Remote job sites often face unreliable GSM signals, slow satellite fallback, and inconsistent Wi-Fi coverage.



- 02 Data Overload:** Machines generate large volumes of unprocessed telemetry and alarm data. Without efficient filtering and prioritization, this data overwhelms cloud systems and operators, leading to alarm fatigue and delays in decision-making.

- 03 Integration Gaps:** Legacy machines and systems from different manufacturers lack a unified communication framework, complicating integration into predictive maintenance and workflow applications.



Solution

Software-Defined Networking (SDN):

Centralized management of data traffic.

Dynamic switching between Wi-Fi, GSM, and satellite networks based on real-time conditions.

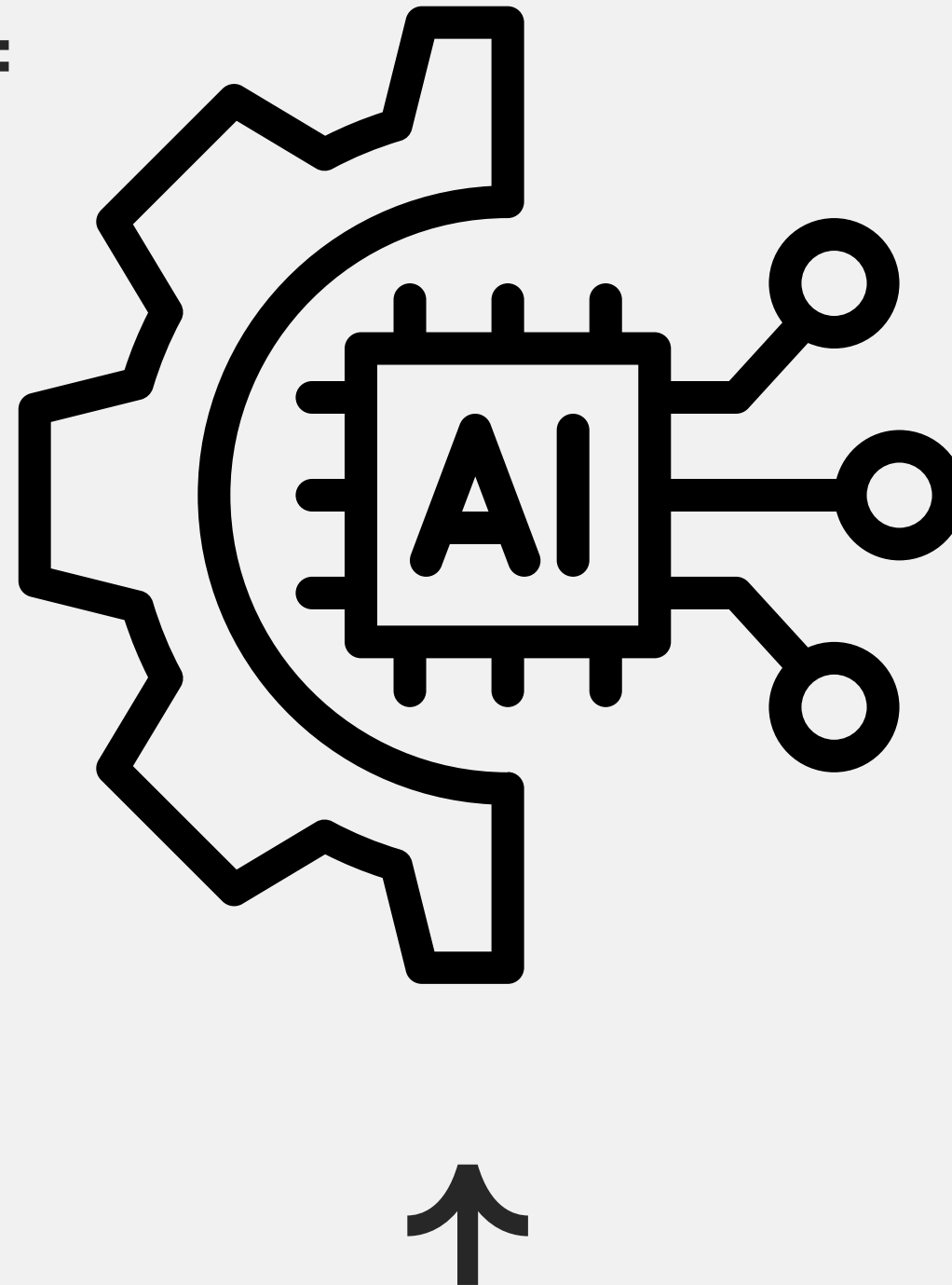
Data prioritization to ensure critical telemetry is transferred with minimal latency.

Edge Computing:

Localized data processing on edge devices (e.g. Raspberry Pi).

Filters and processes raw telemetry data to reduce noise and prioritize actionable insights.

Reduces bandwidth usage by sending only critical alerts or summarized data to the cloud.



Results

with this solution we can improve the Predictive maintenance. and reduce cloud processing load and operator fatigue.

Ensure continuous data updates for connected maps app and more reliable decision making on site